AI Agent Framework Comparison 2026

Comprehensive analysis of 9 open-source AI agent frameworks for self-hosted deployment

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Executive Summary

The AI agent ecosystem exploded in early 2026, with OpenClaw reaching 430K GitHub stars in under 90 days and spawning dozens of alternatives. This study evaluates the 9 most significant open-source AI agent frameworks through a structured analysis of security audits, code reviews, community data, and architectural assessments from over 30 independent sources including Microsoft, Cisco, Kaspersky, NVIDIA, and academic research.

Each agent is scored across 6 weighted dimensions: **Security (30%)**, **Code Quality (20%)**, **Orchestration (20%)**, **Ecosystem (15%)**, **Popularity (10%)**, and **Hardware Efficiency (5%)**. Security is weighted highest because these agents have system-level access, handle credentials, and execute arbitrary code. The study intentionally weights popularity low (10%) to avoid conflating adoption with quality — the most popular agent (OpenClaw, 430K stars) ranks 7th due to 9 CVEs and no built-in sandbox.

Key findings: Rust-based agents (IronClaw, Moltis) lead in security and code quality. OpenHands leads in orchestration API maturity. OpenClaw has the richest ecosystem but the worst security posture. PicoClaw is the most hardware-efficient (10MB RAM) but pre-v1.0. All scores, data points, and claims are traceable to the sources listed at the end of this document.

Comparison Table

SCORE	AGENT	LANGUAGE	STARS	LICENSE	SECURITY	CODE QUALITY	ORCHESTRATION	ECOSYSTEM	POPULARITY	HARDWARE	MIN HW	DOCKER IMAGE	PORT
7.5	**IronClaw**	Rust	4.8K+	Apache 2.0	9/10 Very High	9/10 Excellent	7/10 Good	6/10	4/10 Growing	6/10 Light	2C/4T, 2GB, 500MB	`nearai/ironclaw`	18789
7.3	**OpenHands**	Python	65K+	MIT	7/10 High	7/10 Good	8/10 Excellent	8/10	8/10 Very High	3/10 Heavy	4C/8T, 8GB, 10GB	`ghcr.io/openhands/openhands`	3000
6.9	**Moltis**	Rust	2.3K+	MIT	7/10 High	9/10 Excellent	6/10 Fair	7/10	3/10 Emerging	8/10 Very Light	1C/2T, 512MB, 100MB	`moltis/moltis`	8080
6.3	**NanoClaw**	Node.js	21.5K+	MIT	7/10 High	6/10 Fair	7/10 Good	5/10	6/10 Notable	5/10 Moderate	2C/4T, 4GB, 5GB	`qwibitai/nanoclaw`	Varies

SCORE	AGENT	LANGUAGE	STARS	LICENSE	SECURITY	CODE QUALITY	ORCHESTRATION	ECOSYSTEM	POPULARITY	HARDWARE	MIN HW	DOCKER IMAGE	PORT
6.2	**NemoClaw**	Node.js	4.6K+	NVIDIA	9/10 Very High	5/10 Fair	6/10 Fair	5/10	4/10 Growing	3/10 Heavy	4C/4T, 8GB, 5GB	nvidia/nemoclaw	18789
6.1	**Goose**	Rust + Python	29K+	Apache 2.0	5/10 Medium	7/10 Good	5/10 Fair	9/10	7/10 High	3/10 Heavy	4C/8T, 8GB, 2GB	block/goose	3000
6.0	**OpenClaw**	Node.js	430K+	MIT	2/10 Low	4/10 Poor	9/10 Excellent	10/10	10/10 Dominant	5/10 Moderate	2C/4T, 4GB, 2GB	alpine/openclaw	18789
5.7	**PicoClaw**	Go	25K+	MIT	5/10 Medium	6/10 Fair	5/10 Fair	5/10	7/10 High	10/10 Minimal	1C/1T, 10MB, 50MB	sipeed/picoclaw	18789
5.4	**NanoBot**	Python	26.8K+	Open Source	5/10 Medium	5/10 Fair	6/10 Fair	4/10	7/10 High	8/10 Very Light	1C/2T, 512MB, 200MB	hkuds/nanobot	18790

Agent Profiles

IronClaw 7.5/10

Language: Rust **Stars:** 4.8K+ **License:** Apache 2.0 **Repo:** github.com/nearai/ironclaw

OpenClaw-inspired rewrite in Rust by Illia Polosukhin, co-author of the Transformer paper ("Attention Is All You Need") / NEAR AI. All untrusted tools run in WebAssembly sandboxes with capability-based permissions. Gateway API compatible with OpenClaw. Zero outbound API calls possible — fully air-gapped operation.

Security: Wasm isolation (seL4-inspired). Zero-default access. Endpoint allowlisting, credential injection, leak detection. FIPS certification path.

Code Quality: Pure Rust, zero unsafe code. Wasm formal verification. Capability-based security model throughout.

Orchestration: OpenClaw-compatible Gateway API, session management, MCP tools, capability-based task control.

Ecosystem: OpenClaw-compatible: MCP, multi-provider (Anthropic, OpenAI, Gemini, Mistral, Ollama), Wasm plugins, REST API.

Pro: Best security (Wasm sandbox, zero-trust), air-gap capable, Rust memory safety

Con: ~5% of OpenClaw skill library, 15ms overhead per skill, higher install friction

Min Hardware: 2 CPU cores, 4 threads, 2 GB RAM, 500 MB storage

OpenHands 7.3/10

Language: Python **Stars:** 65K+ **License:** MIT **Repo:** github.com/OpenHands/OpenHands

AI-driven software development platform. Best-in-class REST API + WebSocket server. Docker-native with event-sourced state model. Raised $18.8M. Supports defining agents in code, running locally or scaling to 1000s in the cloud. Enterprise-ready with VPC deployment.

Security: Docker container isolation. Event-sourced state for audit/replay. Fine-grained access control.

Code Quality: Event-sourced, deterministic replay, SWE-bench tested. Python (no memory safety guarantees).

Orchestration: Full REST+WS API, Agent SDK for composing agents, multi-agent support, CI/CD integration, health endpoints, task abort.

Ecosystem: Enterprise: REST+WS API, Agent SDK, MCP, Docker runtime, event sourcing, K8s deploy.

Pro: Best REST API, Docker isolation, $18.8M funded, 87% bug tickets solved same day

Con: Resource-heavy (8GB+ RAM), looping on ambiguous tasks, high LLM API costs

Min Hardware: 4 CPU cores, 8 threads, 8 GB RAM, 10 GB storage

Moltis 6.9/10

Language: Rust **Stars:** 2.3K+ **License:** MIT **Repo:** github.com/moltis-org/moltis

Single Rust binary — no Node.js, no npm, no runtime dependencies. Runs on Mac Mini, Raspberry Pi, or any server. Built-in voice, memory, scheduling, Telegram, Discord, browser automation, MCP servers. Streaming-first responses. Keys never leave your machine.

Security: Container sandbox. No telemetry, no phone-home. Full audit trail. Keys never leave machine.

Code Quality: 150K lines Rust, zero unsafe workspace-wide, 2,300+ tests, Prometheus/OTel, auditable agent loop (~5K lines).

Orchestration: JSON-RPC API, built-in sessions, memory system for context, cron scheduling, MCP tools, health endpoint.

Ecosystem: Built-in: MCP, voice, memory, scheduling, Telegram, Discord, browser, 8 TTS/7 STT providers.

Pro: Single binary, no runtime deps, voice+memory+MCP built-in, runs on Raspberry Pi

Con: Small community (2.3K stars), limited channel support, integration/tuning overhead

Min Hardware: 1 CPU core, 2 threads, 512 MB RAM, 100 MB storage

NanoClaw 6.3/10

Language: Node.js (TypeScript) **Stars:** 21.5K+ **License:** MIT **Repo:** github.com/qwibitai/nanoclaw

Container-first agent built on Claude Agent SDK by Qwibit AI. Every agent session runs in an isolated Docker container. Supports "Agent Swarms" — teams of Claude instances collaborating in parallel. Docker partnership announced March 2026.

Security: OS-level container isolation per session. "Isolation over defense" — hard limits at container layer. Provable, auditable security.

Code Quality: 700 lines TypeScript. Tiny footprint. Container isolation compensates for minimal app-layer security. 62 contributors.

Orchestration: Container-per-session, Agent Swarms (parallel Claude instances), Docker Sandbox integration, multi-channel, container lifecycle.

Ecosystem: Focused: Claude SDK, Agent Swarms, Docker isolation, multi-channel, memory.

Pro: OS-level container isolation per session, Docker partnership, Agent Swarms

Con: Requires Anthropic API key (Claude only), heavier than lightweight alternatives

Min Hardware: 2 CPU cores, 4 threads, 4 GB RAM, 5 GB storage

NemoClaw 6.2/10

Language: Node.js + OpenShell **Stars:** 4.6K+ **License:** NVIDIA **Repo:** github.com/NVIDIA/NemoClaw

NVIDIA enterprise security layer for OpenClaw. Announced at GTC 2026. Triple enforcement: Sandbox (Landlock + seccomp + network namespace), Policy Engine (filesystem/network/process rules), Privacy Router (inference routing + PII stripping). 85.6% PinchBench score.

Security: Landlock + seccomp + network NS. Policy engine + PII router. Enterprise-grade by NVIDIA.

Code Quality: Alpha stage. NVIDIA backing. "Expect rough edges." Not production-ready per NVIDIA's own documentation.

Orchestration: Via OpenClaw Gateway (inherits full API), policy-filtered context, OpenShell health monitoring, privacy router.

Ecosystem: Security stack: OpenShell sandbox, policy engine, privacy router, via OpenClaw ecosystem.

Pro: Triple enforcement (Landlock+seccomp+NS), PII router, NVIDIA backing, 85.6% PinchBench

Con: Alpha/not production-ready, Linux only, missing fleet management, limited model support

Min Hardware: 4 CPU cores, 4 threads, 8 GB RAM, 5 GB storage

Goose 6.1/10

Language: Rust + Python **Stars:** 29K+ **License:** Apache 2.0 **Repo:** github.com/block/goose

By Block (Square/Cash App/Afterpay). Extensible AI agent with CLI and Electron desktop interfaces. Connects to 3,000+ tools through MCP protocol. Automates workflows with "recipes." Model-agnostic. 400+ contributors. 60% of Block workforce uses it weekly.

Security: No built-in sandbox. Red-teamed by Block security ("Operation Pale Fire"). Found and fixed prompt injection via invisible Unicode.

Code Quality: Rust core, red-teamed by Block security team, 400+ contributors, mixed Rust+Python codebase.

Orchestration: Extension protocol, CLI-focused (not API-first), recipes for workflows, MCP tools, no native sub-agents.

Ecosystem: Extensive: 3K+ MCP tools, recipes, extensions, Electron desktop, plugin system, grant program.

Pro: 3K+ MCP tools, Block backing, 50-75% dev time savings reported, free, model-agnostic

Con: No built-in sandbox, prompt injection found (red team), terminal-centric learning curve

Min Hardware: 4 CPU cores, 8 threads, 8 GB RAM, 2 GB storage

OpenClaw 6.0/10

Language: Node.js **Stars:** 430K+ **License:** MIT **Repo:** github.com/openclaw/openclaw

The original open-source personal AI assistant. Fastest-growing open-source project in history (220K stars in 84 days). 2M monthly active users, 27M web visits/month. Built-in HTTP Gateway API. Connects to WhatsApp, Telegram, Discord, Slack. Multi-provider: Anthropic, OpenAI, Ollama.

Security: None built-in. No permission model, no code signing, full FS/shell/network access. 9 CVEs including CVSS 8.8 RCE (CVE-2026-25253). 341 malicious skills on ClawHub. 40K+ exposed instances.

Code Quality: 430K lines. Security score 28/100. 111 red flags. $373K technical debt. Poor test coverage (526h remediation needed).

Orchestration: Full REST+WS Gateway, session mgmt (thread modes), native sub-agents, agent teams with handoff protocols, cron, webhooks, task abort.

Ecosystem: Massive: 10K+ skills, ClawHub marketplace, MCP (3K+), webhooks, cron, 5+ channels, REST+WS API.

Pro: Largest ecosystem (10K+ skills), most integrations, 2M active users, battle-tested

Con: 9 CVEs incl. RCE, no sandbox, 341 malicious skills on ClawHub, credential leaks

Min Hardware: 2 CPU cores, 4 threads, 4 GB RAM, 2 GB storage

PicoClaw 5.7/10

Language: Go **Stars:** 25K+ **License:** MIT **Repo:** github.com/sipeed/picoclaw

Ultra-lightweight Go agent by Sipeed. Under 10MB RAM, boots in 1 second. Runs on $5/$10 hardware. 99% less memory than OpenClaw. Rebuilt from the ground up in Go through a "self-bootstrapping" process. Native MCP integration, vision pipeline for multimodal LLMs.

Security: Cron security gating. Pre-v1.0 — "unresolved security issues." Not for production-critical workloads yet.

Code Quality: Go, lowest issue ratio (0.013), but 95% AI-generated code. Pre-v1.0. 187 contributors.

Orchestration: Gateway API (OpenClaw-like), basic sessions, spawn_status for sub-agents, cron, MCP.

Ecosystem: Growing: MCP, vision pipeline, sub-agents, cron, system tray UI, multimodal.

Pro: 10MB RAM, boots in 1s, runs on $5 hardware, single binary, vision pipeline

Con: Pre-v1.0 security caveats, limited complex workflows, basic state management

Min Hardware: 1 CPU core, 1 thread, 10 MB RAM, 50 MB storage

NanoBot 5.4/10

Language: Python **Stars:** 26.8K+ **License:** Open Source **Repo:** github.com/HKUDS/nanobot

Ultra-lightweight AI assistant from HKU Data Intelligence Lab. ~4,000 lines of Python — 99% smaller than OpenClaw. Gateway mode with HTTP API. Supports subagent spawning for parallel tasks. "Auditable in an afternoon."

Security: Minimal sandboxing. Small codebase (4K lines) naturally limits attack surface. Relies on LLM provider security.

Code Quality: 4K lines Python, auditable in an afternoon, minimal tests, 215 contributors.

Orchestration: Gateway HTTP API, basic sessions, native subagent spawning, multi-provider, channel adapters.

Ecosystem: Minimal: Gateway API, subagent spawning, multi-provider, channel adapters.

Pro: Only 4K lines, easiest to understand/hack, multi-provider, subagent spawning

Con: Minimal sandbox, small ecosystem, limited advanced features compared to OpenClaw

Min Hardware: 1 CPU core, 2 threads, 512 MB RAM, 200 MB storage

Score Methodology

DIMENSION	WEIGHT	WHAT IT MEASURES
Security	30%	Sandboxing approach, known CVEs, credential handling, audit results, zero-trust architecture
Code Quality	20%	Language memory safety, test coverage, technical debt, code audits, unsafe code blocks
Orchestration	20%	API programmability, session management, sub-agent support, task control, multi-agent coordination
Ecosystem	15%	Tools, plugins, skills, MCP integrations, channels, protocols, marketplace
Popularity	10%	GitHub stars, community size, media coverage, corporate backing, enterprise adoption
Hardware	5%	Resource efficiency — inverse of minimum RAM/CPU requirements

Why these weights? Security receives the highest weight (30%) because AI agents have system-level access — they execute shell commands, read files, handle API keys, and interact with external services. A compromised agent is a compromised machine. Code Quality and Orchestration each receive 20% because they directly determine reliability and controllability. Ecosystem matters (15%) but is secondary to safety. Popularity is deliberately low (10%) to prevent conflating adoption with quality — OpenClaw demonstrates this clearly: 430K stars but 9 CVEs and no sandbox. Hardware efficiency is minimal (5%) because most deployments have adequate resources; it only matters for edge/IoT use cases.

Limitations: This analysis reflects the state of these projects as of March 2026. Open-source projects evolve rapidly. Security scores may change as vulnerabilities are discovered or fixed. Orchestration and ecosystem scores are based on documented features, not exhaustive testing. Performance benchmarks (SWE-bench, PinchBench) measure the underlying LLM model more than the agent framework itself, which is why we excluded them from the scoring formula. All data is sourced from public information; no private audits were conducted.

Sources & References

All scores and claims in this document are traceable to the following sources. Links verified as of March 2026.

Security Audits & Research

- Microsoft Security Blog: Running OpenClaw Safely — Identity, isolation, runtime risk analysis
- Cisco: Personal AI Agents Are a Security Nightmare — Enterprise security assessment
- Kaspersky: Key OpenClaw Risks — Credential exposure and infostealer targeting
- Block Red Team: Operation Pale Fire — Goose prompt injection via invisible Unicode
- Bitsight: OpenClaw Security Risks — 40,214 exposed instances, 35.4% vulnerable
- IronClaw Security Review — Wasm sandbox architecture and seL4-inspired capability model

Code Quality & Architecture

- OpenClaw Audit: Modern Core with Hidden Debt — $373K tech debt, security 28/100, 111 red flags

- Moltis: Zero Unsafe, Full Audit Trail — 150K Rust lines, 2,300+ tests, zero unsafe
- Rust Agent Runtime Showdown — Moltis vs ZeroClaw architecture comparison
- OpenHands Software Agent SDK — Event-sourced architecture academic paper
- 116K Stars in 8 Weeks: Four Teams Rewrote OpenClaw — Codebase size and contributor comparisons

Community & Popularity

- OpenClaw Statistics 2026 — 430K stars, 2M MAU, 27M web visits, 900+ contributors
- Goose AI Review 2026 — 29K stars, 400+ contributors, Block backing
- NanoClaw + Docker Partnership (TechCrunch) — Docker deal, community growth
- OpenHands Review 2026 — 65K stars, $18.8M funding, enterprise adoption
- PicoClaw GitHub — 25K stars, edge/IoT community

Comparisons & Analysis

- Full 2026 Comparison: ZeroClaw vs OpenClaw vs NanoClaw vs NanoBot vs PicoClaw vs IronClaw
- OpenClaw Alternatives Compared (AI Magicx)
- NemoClaw at GTC 2026 (TechCrunch)
- Agent Wars 2026
- Security-First Guide to Self-Hosted AI Assistants

Benchmarks

- SWE-bench Verified — Software engineering benchmark
- Terminal-Bench — Terminal agent benchmark
- PinchBench — OpenClaw agent performance benchmark

Forums & Community Discussions

- Hacker News: NanoClaw + Docker Discussion
- Hacker News: Moltis Show HN Launch
- DEV Community: Claw Family Overview
- Product Hunt: IronClaw Launch